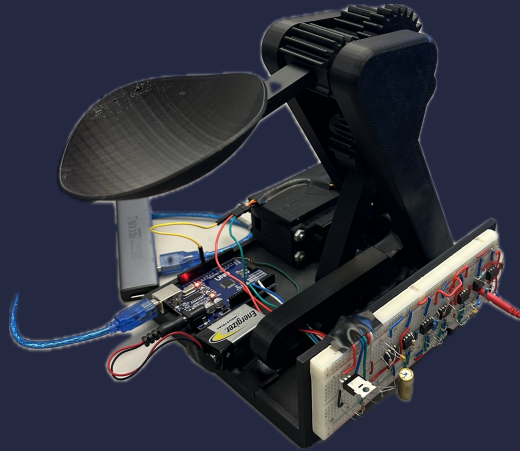


Aim Assist

Have Precision in Every Shot



Raiyan Aaijaz & Mekhael Thaha



Objective and Requirements:

Develop a prosthetic arm capable of shooting a tennis ball-sized object with the technique of a basketball shot. The prosthetic should mimic the biomechanics of a human arm, ensuring consistent release speed, angle, and spin to achieve accurate and repeatable shots.

Torque: 2.57 N-m

Speed: 2.48 m/s

Range of Motion: 53 degree

Engineering Requirements Calculations

Mass of tennis ball (m_1) = 0.057 Kg

estimated mass of prosthetic (m_2) = 0.714 Kg

Length (l) = 0.221 m

time (t) = 0.2 s

$\theta = 58^\circ = 1.012 \text{ rad}$

Angular Acceleration

$$\theta_f = \theta_i + \omega_i t + \frac{1}{2} \alpha t^2$$

$$1.012 = \frac{1}{2} \alpha (0.2)^2$$

$$\frac{2(1.012)}{0.2^2} = \alpha$$

$$\alpha = 50.6 \text{ rad/s}^2$$

Moment of Inertia

$$I = I_1 + I_2$$

$$= \frac{1}{3} m_1 l_1^2 + m_2 l_2^2$$

$$= \frac{1}{3} (0.057)(0.221)^2 + (0.714)(0.221)^2$$

$$= 0.0358 \text{ Kg} \cdot \text{m}^2$$

Torque

$$T_{\text{net}} = T_1 + T_2$$

$$= I \alpha + T_{\text{mg}}$$

$$= 0.0358 (50.6) + 0.7612$$

$$= 2.5728 \text{ N} \cdot \text{m}$$

Torque Due to Weight

$$T_{\text{mg}} = m_1 g r_1 \sin \theta + m_2 g r_2 \sin \theta$$

$$= 0.057 (9.81)(0.221)(\sin 58^\circ) + 0.714 (9.81)(0.1105)(\sin 58^\circ)$$

$$= 0.1046 + 0.6564$$

$$= 0.7612$$

Optimal Speed

$$\text{Optical release speed for basketball} = 8.2 \frac{\text{m}}{\text{s}}$$

$$\text{Approx. Mass of Basketball} = 0.625 \text{ kg}$$

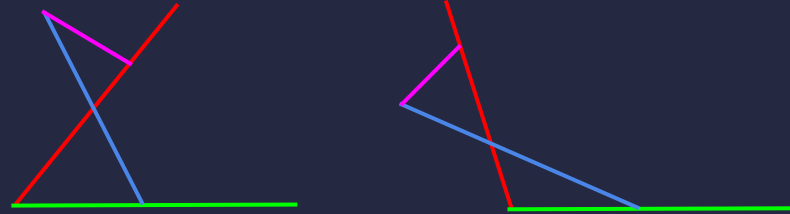
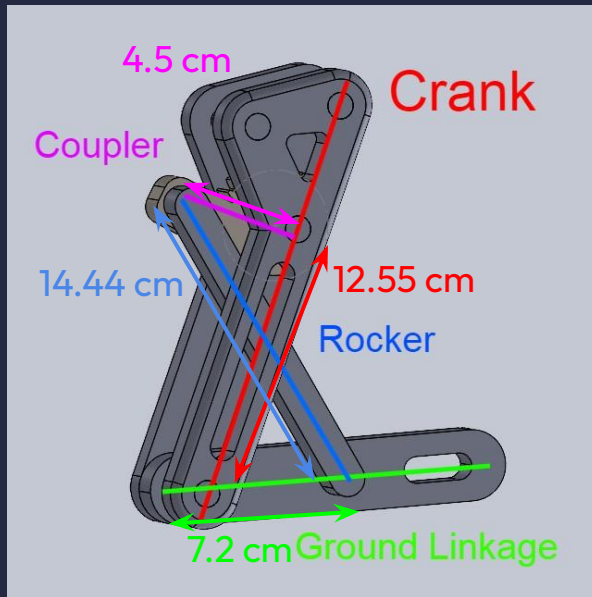
$$\text{Approx. Mass of Tennis Ball} = 0.057 \text{ kg}$$

$$v_{\text{tennis}} = v_{\text{basketball}} \times \sqrt{\frac{m_{\text{tennis}}}{m_{\text{basketball}}}}$$

$$= 8.2 \times \sqrt{\frac{0.057}{0.625}}$$

$$= 2.48 \frac{\text{m}}{\text{s}}$$

Mechanism Design



Grashof Linkage Proof: $S + L < P + Q$

$$\begin{aligned} S + L &= 4.5 + 14.44 \\ &= 18.94 \end{aligned}$$

$$\begin{aligned} P + Q &= 12.55 + 7.2 \\ &= 19.75 \end{aligned}$$

$$\therefore S + L < P + Q$$

Material and Servo Analysis

Material Selection

We chose 3D printing using Polylactic Acid (PLA) for all components due to several advantages:

- **Ease of Manufacture:** The gears designed have large teeth, which eliminates the need for high-precision cutting tools such as laser cutters.
- **Material Properties:** PLA is a lightweight material. When printed with increased infill, it provides the necessary sturdiness for our application.
- **Cost-Effectiveness:** PLA is a cost-effective choice for prototyping, allowing for efficient and economical production of design iterations.

Servo Selection

Chosen Motor: HS-805BB Servo Motor

- Torque requirement needed for motor was 2.57 N-m
- HS-805BB gets 1.95 Nm at 4.8 V, 2.52 Nm at 6.0 V
- Though it is the heaviest option at 152 grams, it is manageable. Lighter motors are inadequate in other areas.

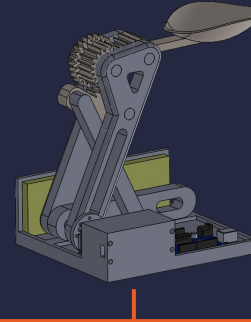


Design Process and Iterations



Iteration 1:

- Motor was planned to be connected at the linkage directly rather than the arm.
- The arms were kept stationary.
- Hand design was much larger than needed.



Iteration 2:

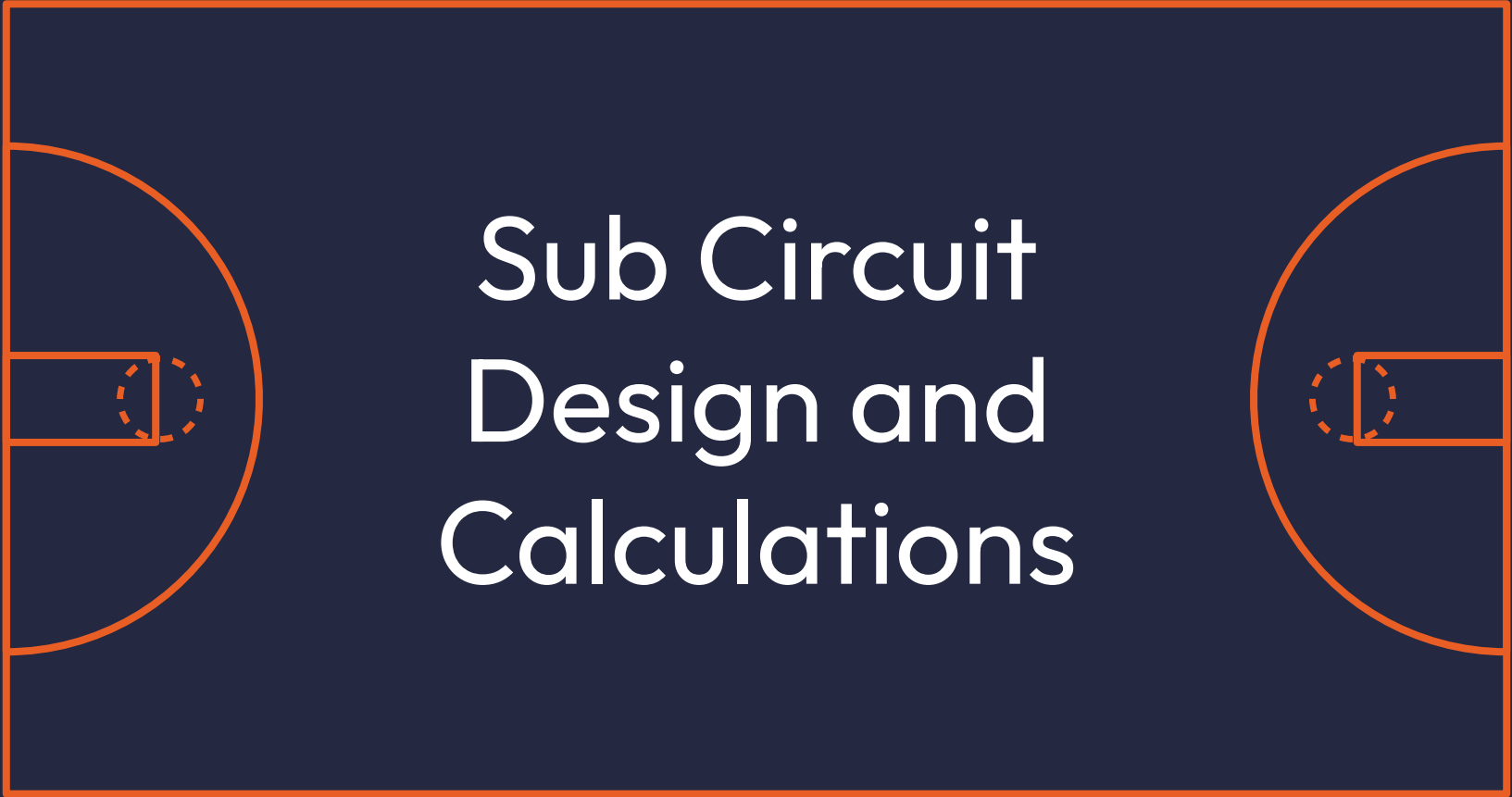
- Created a base to store Arduino, breadboard, and motor.
- Improved the hand design
- Moved the motor to add additional motion
- Scaled the design by 80% to minimize the weight.

Arduino Code

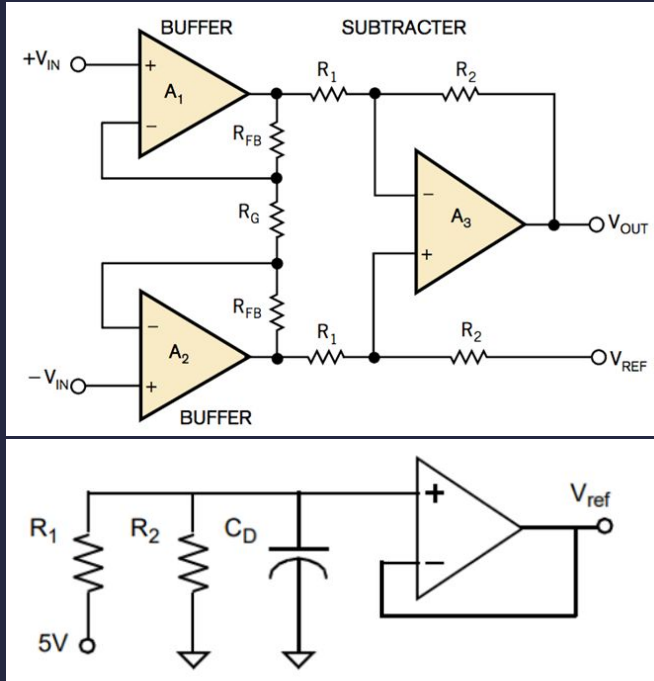
```
1  #include <Servo.h>
2
3  Servo Motor;
4
5  int signalPin = A0; //analog pin connected to the circuit output
6  int motorControlPin = 9; //digital ~ pin connected to the motor control wire
7  int motorPosition = 0;
8  int desiredAngle = 90;
9
10 void setup() {
11
12     Serial.begin(9600);
13     pinMode(motorControlPin, OUTPUT); //setting motor control pin as output
14     digitalWrite(motorControlPin, LOW); //setting motor to be off initially
15
16     Motor.attach(motorControlPin); //attaching the motor to the motor control pin
17     Motor.write(motorPosition); //starting position of the motor
18     Serial.begin(9600);
19 }
```

```
21 void loop() {
22
23     int signalValue = analogRead(signalPin); //read the signal
24     Serial.println(signalValue);
25
26
27     Motor.write(motorPosition);
28
29
30
31     if (signalValue > 670 && motorPosition == 0){
32         Serial.println(1);
33         delay(750);
34         motorPosition = desiredAngle;
35         Motor.write(motorPosition);
36
37         delay(1000);
38         motorPosition = 0;
39         Motor.write(motorPosition);
40         delay(5);
41
42     }
43 }
```

Sub Circuit Design and Calculations



Instrumentation Amplifier + Virtual Ground



Responsible for recognizing the EMG signal produced and eliminating noise from the skin.

- V_{ref} was set using a voltage divider and follower stage
- More accurate V_{ref} for longer periods of time, for V values and used to eliminate negative signal

R_{gain} Calculations:

Desired Gain = 1000

$$R_1 = R_2 = R_{FB} = 50 \text{ k}\Omega$$

$$A_v = 1 + \frac{2R}{R_{gain}}$$

$$1000 = 1 + \frac{2(50000)}{R_{gain}}$$

$$R_{gain} = \frac{100000}{999}$$

$$= 100.1 \Omega$$

$$R_{gain} \approx 100 \Omega$$

V_{ref} Calculations:

$$R_1 = R_2 = 500 \Omega \text{ (Chosen value)}$$

$$V_{ref} = V_{in} \cdot \frac{R_2}{R_1 + R_2}$$

$$= 5 \cdot \frac{500}{1000}$$

$$= 2.5$$

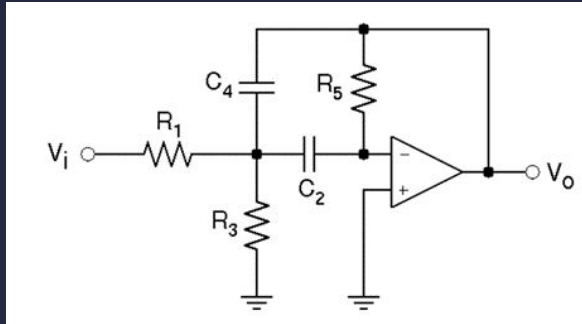
$$C_D = \frac{\tau}{R_{eq}} = 1 \mu F$$

$$R_{eq} = R_1 || R_2 = 250 \Omega$$

BandPass Filter (20 Hz and 500 Hz)

Eliminates excess noise from the signal and ensures only EMG signal is passed.

- Theoretical cutoffs were selected based on the estimated frequency range of EMG signals for muscle contraction.



$$C_2 = 1 \mu f$$

$$C_4 = 33 nf$$

$$f_o = \frac{20 + 500}{2} + 20$$

$$= 280 Hz$$

$$\Delta F = 500 - 20$$

$$= 480 Hz$$

$$Q = \frac{f_o}{\Delta F}$$

$$= \frac{280}{480}$$

$$= 0.5833$$

$$R_1 = \frac{Q}{G2\pi f_o C_4}$$

$$= \frac{0.5833}{1(2\pi)(280)(33 \times 10^{-9})}$$

$$= 10047.1$$

$$\approx 10 k \Omega \text{ (Chosen Value)}$$

$$R_5 = \frac{Q(C_2 + C_4)}{2\pi f_o C_2 C_4}$$

$$= \frac{0.5833(33 \times 10^{-9} + 1 \times 10^{-6})}{(2\pi)(280)(33 \times 10^{-9})(1 \times 10^{-6})}$$

$$= 10378.6$$

$$\approx 10 k \Omega \text{ (Chosen Value)}$$

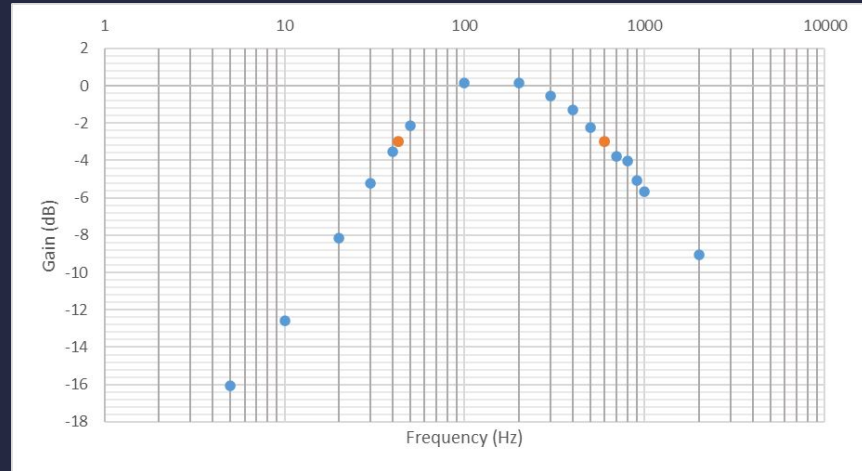
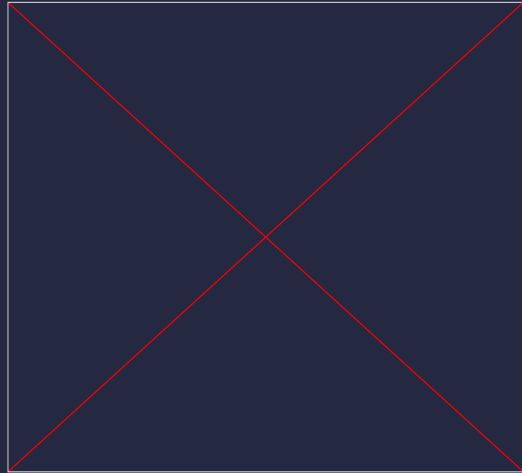
$$R_3 = \frac{Q}{(2\pi f_o)^2 R_5 C_2 C_4 - \frac{1}{R_1}}$$

$$= \frac{0.5833}{(2\pi(280))^2 (10378.6)(1 \times 10^{-6})(33 \times 10^{-9}) - \frac{1}{10047.1}}$$

$$= 607.27$$

$$\approx 620 \Omega \text{ (Chosen Value)}$$

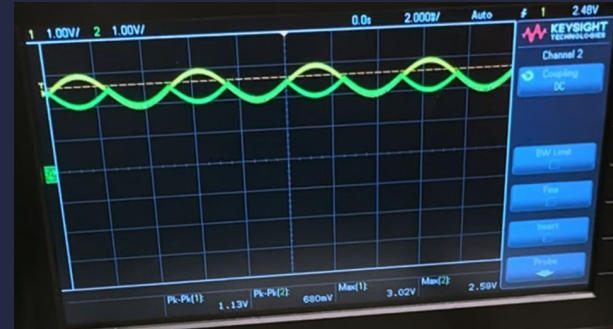
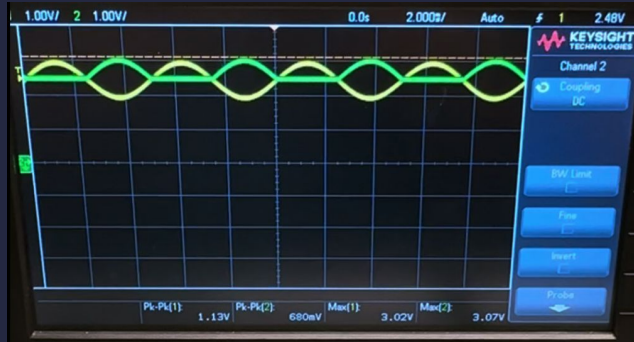
BandPass Filter Video + Bode Plot



Full-Wave Rectifier

Responsible for cleaning up signal quality and emphasizing peaks of the EMG signal while maintaining integrity of the signal

- Gain of -1
 - 2 from half wave rectifier
 - +1 from summing amplifier



For $0 < t < T/2$:

$$V_1 = 0.680V$$

$$V_2 = -\frac{1.13}{2} = -0.565V$$

$$V_0 = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right)$$

$$= -20K \left(\frac{0.68}{10K} - \frac{0.565}{20K} \right)$$

$$= -0.795$$

$$\frac{V_0}{V_{in}} = \frac{-0.795}{1.13}$$

$$= -0.704$$

$$\text{Total gain} = -1.204 \approx -1$$

For $T/2 < t < T$

$$V_1 = 0V$$

$$V_2 = \frac{1.13}{2} = 0.565V$$

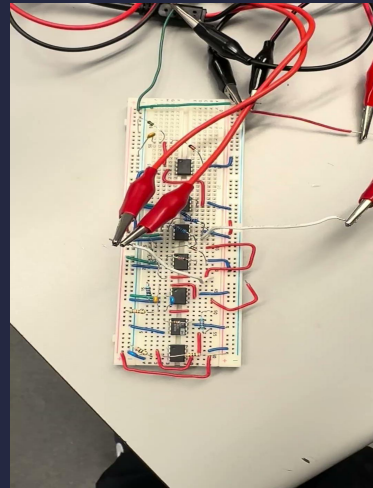
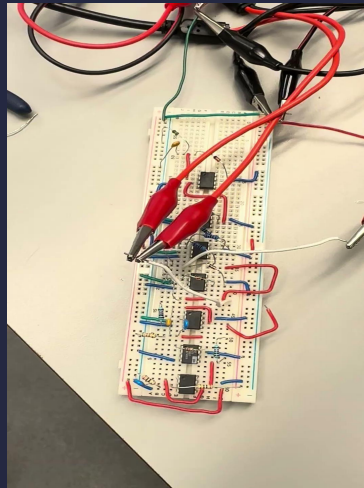
$$V_0 = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right)$$

$$= -20K \left(0 + \frac{0.565}{20K} \right)$$

$$= -0.565$$

$$\frac{V_0}{V_{in}} = \frac{-0.565}{1.13} = -0.5$$

Full-Wave Rectifier Videos



Inverting Amplifier

Responsible for inverting the peaks from the rectifier (positive peaks) and adding a gain to make peaks easier to read.

- Gain of 2.5 was selected.
- Note that picture shows a gain of 1, however the R_F value was updated later.



$$V_{out} = V_{in} \left(-\frac{R_F}{R_{IN}} \right)$$

$$\frac{V_{out}}{V_{in}} = -\frac{R_F}{R_{IN}} = Gain$$

$$Gain = -\frac{5000}{2000} \\ = 2.5$$



Inverting Amplifier Videos



Peak Detector

Responsible for identifying the peaks of the signal corresponding to muscle contractions and holding for a fixed amount of time before searching for the new peak

- Allows voltages to be read once a threshold voltage of ~ 3.5 is achieved.
- Triggers the mechanism to contract after a short delay.
- R was added to allow for the peaks to hold for longer (takes longer for capacitor to drain out)

Time Constant Calculations:

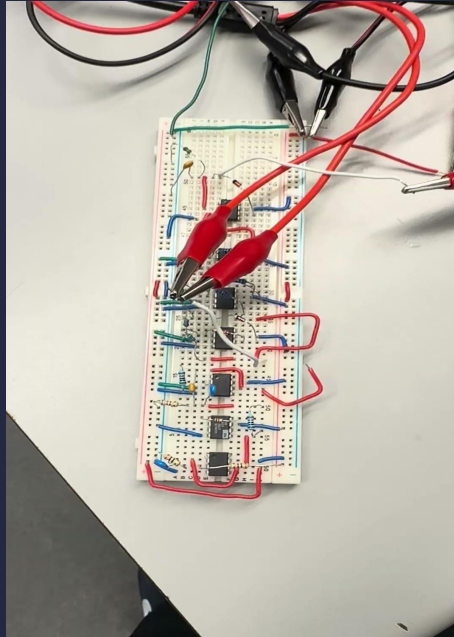
$$\tau = RC$$

$$\tau = 1 \times 10^6 (47 \times 10^{-6})$$

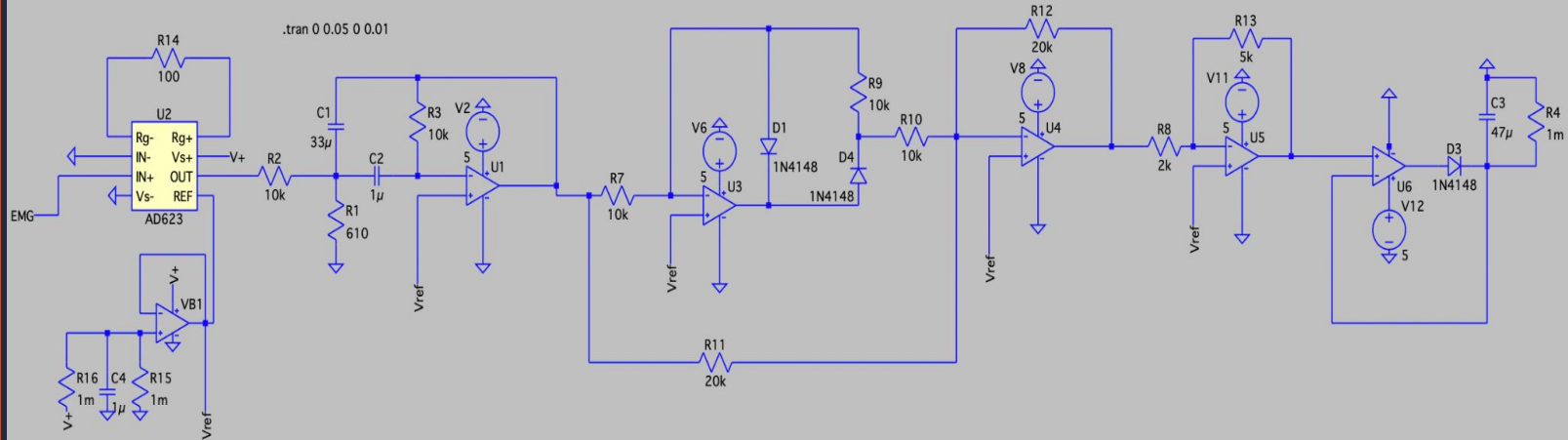
$$\tau = 47s$$

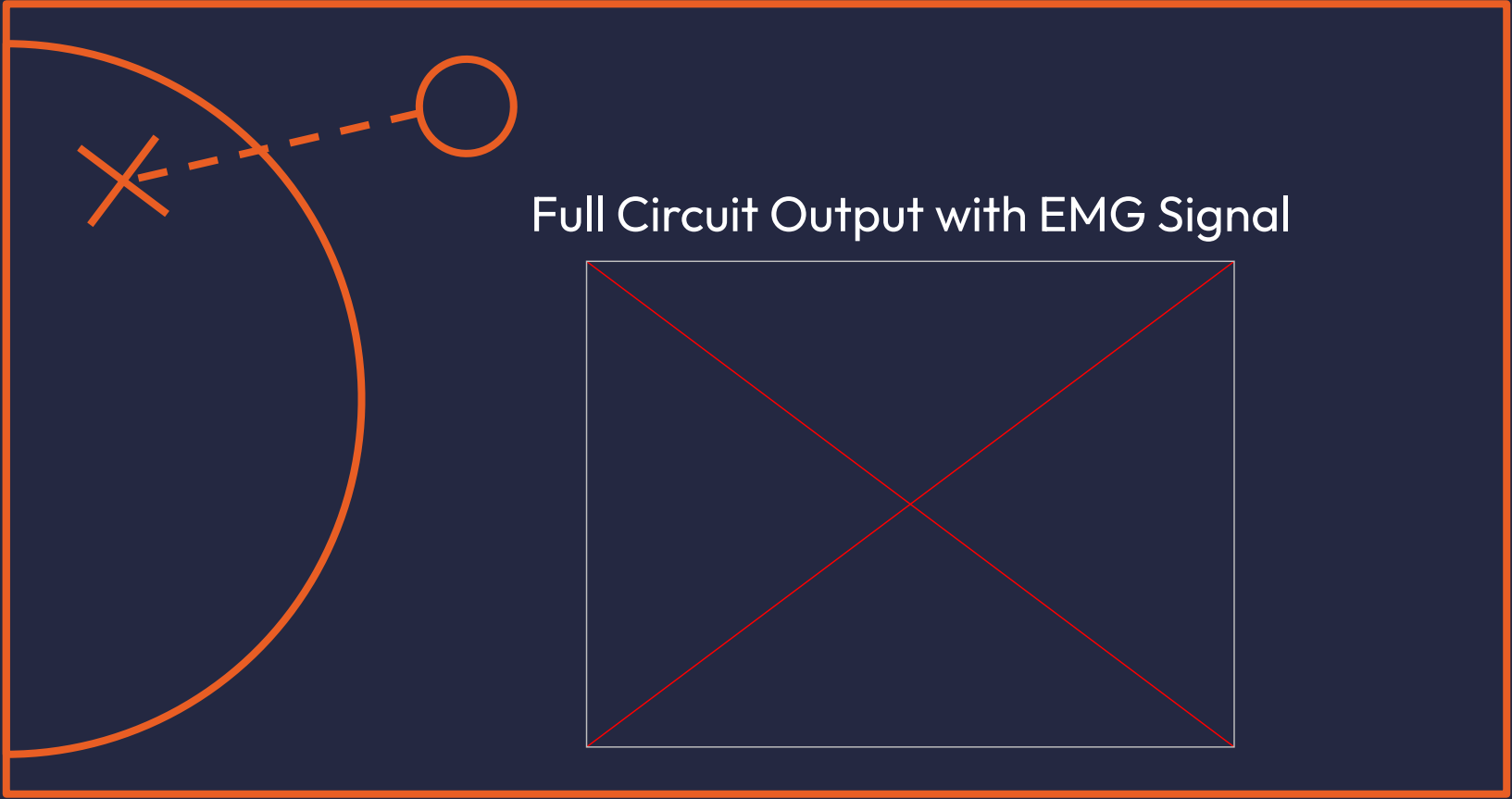


Peak Detector Video



Final Circuit Schematic





Finish Product Demo

